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Development of a Pregnancy-Specific Physiologically Based Pharmacokinetics (PBPK) Model for Aspirin

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ABSTRACT

Aspirin is one of the most commonly used medications in pregnancy, particularly for the prevention of hypertensive disorders. Despite aspirin's widespread use in pregnancy for preeclampsia prevention, its pharmacokinetics (PK) across all trimesters remain poorly characterized, complicating optimal dosing recommendations. To develop a pregnancy-specific physiologically based pharmacokinetic (PBPK) model for aspirin that could be individualized to patient-specific parameters, illustrating differences in aspirin PK across the different trimesters of pregnancy. A PBPK model was developed using GastroPlus (a mechanistically driven simulation software) for nonpregnant and pregnant people at each trimester of pregnancy. The nonpregnant PBPK model was first established and validated against existing data from healthy adult volunteers. Once validated, the model was adapted for pregnant people and verified using observed pharmacokinetic profiles. The simulated PK parameters of aspirin in pregnant and nonpregnant women closely matched the clinical observations reported in the literature, with fold errors ≤ 1.04 (less than 1.5 is considered an acceptable simulation model). The predicted systemic exposure (AUC_{0-24h}) of salicylic acid (SA), the active metabolite of aspirin decreased throughout gestation, showing a reduction of approximately 20% at 10 weeks and 30% at 40 weeks. An increase in clearance was observed as gestation progressed. The model predicted a modest decrease of 10% in systemic exposure in pregnant women and a 20% increase in fetal exposure to SA as pregnancy progresses. A PBPK model using GastroPlus was developed to describe the PK and pharmacodynamics of aspirin in both pregnant and nonpregnant healthy adults.

1 | Introduction

Hypertensive disorders of pregnancy (HDP) contribute significantly to maternal morbidity and mortality, affecting 1 in 7 births in the United States, and are considered one of the leading causes of pregnancy-related death in the United States [1, 2]. HDP, particularly preeclampsia, is thought to

evolve from abnormal early placental development that results in placental ischemia and angiogenic imbalance, leading to systemic inflammation, endothelial dysfunction, and oxidative stress [3]. On the other hand, aspirin (acetylsalicylic acid), a commonly used nonsteroidal anti-inflammatory drug with pro-angiogenic, anti-inflammatory, and antiplatelet properties, is believed to contribute to its preventive effect on HDP [4]. Aspirin

Ana Collins-Smith and Ananth Kumar Kammala shared first authorship.

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Study Highlights

- What is the current knowledge on the topic?
 - Aspirin is commonly used during pregnancy, particularly to prevent preeclampsia. However, limited research exists on aspirin's pharmacokinetics (PK) in pregnancy, with most studies only covering specific trimesters. The optimal dosing strategy for pregnant women remains unclear, leading to ongoing debates in obstetrics. Salicylic acid (SA), the active metabolite of aspirin, undergoes significant pharmacokinetic changes during pregnancy, but these have not been fully characterized across all trimesters.
- What question did this study address?
 - This study aimed to characterize the pharmacokinetics of aspirin across all trimesters of pregnancy using a PBPK model. It sought to determine whether dose adjustments are necessary as pregnancy progresses. The study also explored fetal and amniotic fluid exposure to aspirin metabolites over gestation.
- What does this study add to our knowledge?
 - The study provides a trimester-specific PBPK model of aspirin, illustrating how maternal and fetal drug exposure changes over time. It shows that maternal plasma concentrations of salicylic acid decrease with gestational age, whereas fetal plasma and amniotic fluid concentrations increase. These findings suggest that current fixed-dose aspirin regimens may be inadequate for maintaining optimal maternal therapeutic levels in later pregnancy. The model supports the need for individualized aspirin dosing based on gestational age.
- How might this change drug discovery, development and/or therapeutics?
 - These findings demonstrate how pregnancy-specific PBPK modeling can accelerate drug discovery and development by enabling early prediction of maternal-fetal exposure profiles. The approach informs clinical development by guiding trimester-specific dosing strategies and optimizing trial design for pregnant populations. From a therapeutic perspective, the model supports personalized medicine in pregnancy, offering a framework to improve maternal and fetal outcomes while reducing the need for extensive clinical studies in this vulnerable population.

(acetylsalicylic acid) is a nonsteroidal anti-inflammatory drug commonly used as an antipyretic, anti-inflammatory, antithrombotic, and analgesic. Currently, it is the only approved agent for the prevention of HDP [5]. Acetylsalicylic acid is an irreversible cyclooxygenase (COX) inhibitor, suppressing prostaglandin and thromboxane production [6].

The oral bioavailability of non-enteric-coated aspirin is approximately 40%–50%, while enteric-coated aspirin exhibits reduced bioavailability due to delayed dissolution and absorption [7]. Orally administered aspirin is rapidly absorbed in the upper gastrointestinal tract and stomach, leading to measurable

inhibition of platelet function within 60 min [8]. Aspirin is primarily absorbed as acetylsalicylic acid; however, about 70% of it is hydrolyzed by liver enzymes and plasma esterases into salicylic acid (SA), its active metabolite [9]. Peak plasma levels occur 30–40 min after ingesting non-enteric-coated aspirin, whereas reaching peak plasma levels can take up to 4 h with enteric-coated formulations. The half-life of aspirin is 15–20 min. Approximately 50%–80% of aspirin in the blood is bound to plasma proteins [5]. As much as 80% of aspirin is metabolized in the liver, with the enzyme CYP2C9 playing the predominant role. The kidneys primarily excrete SA with an elimination half-life of 2.0–2.4 h (in doses less than 250 mg) [10]. During pregnancy, numerous temporary physiological, anatomical, and metabolic changes can affect the pharmacodynamics (PD) and pharmacokinetics (PK) of medications like aspirin (noted in Table 1). In several gestational-age-specific clinical studies (first and third trimester), aspirin PK were determined [22–24]. However, studies assessing the PK and PD of aspirin in all three trimesters are lacking due to difficulty completing PK and PD studies throughout pregnancy. Furthermore, while it is assumed changes in the expression and activity of specific metabolic enzymes and membrane transport proteins during pregnancy drive these changes, the specific mechanisms driving the time-varying PK of aspirin have not been assessed. To address the gaps in PK knowledge during gestation, we developed a pregnancy physiological-based pharmacokinetic (PBPK) model using GastroPlus incorporating the essential physiologic and metabolic changes that occur during pregnancy. This model may be particularly useful to researchers as it provides a better understanding of the biodisposition of aspirin during pregnancy, facilitating improved exposure-response assessments for safety and efficacy. By predicting both the maternal and fetal PK of aspirin throughout pregnancy, this model will eventually help healthcare providers tailor aspirin therapy to minimize risks and maximize benefits for pregnant women, thereby improving outcomes for both the mother and the fetus.

Furthermore, while it is assumed that changes in the expression and activity of specific metabolic enzymes and membrane transport proteins during pregnancy drive these pharmacokinetic differences, the specific mechanisms underlying the time-varying PK of aspirin have not been fully assessed. To address these gaps, we developed a pregnancy physiologically based pharmacokinetic (PBPK) model using GastroPlus that incorporates the essential physiologic and metabolic changes occurring during gestation. The pregnancy PBPK model incorporated plasma volume expansion, increased cardiac output and renal clearance, changes in gastrointestinal motility, and pregnancy-induced modulation of hepatic CYP450 enzymes (CYP2C9, CYP3A4, CYP2E1), UGTs, and esterase parameters selected based on their established relevance to aspirin absorption, metabolism, and elimination. In addition, fetal-specific parameters were integrated, including gestational changes in placental blood flow, placental volume, and permeability, as well as fetal plasma protein binding, blood volume, and tissue partition coefficients. Amniotic fluid accumulation and clearance were modeled as a dynamic compartment increasing across gestation, and consistent with the literature, fetal metabolism and clearance were assumed to be minimal, with maternal-placental transfer serving as the primary elimination pathway. This model may be beneficial to researchers as it provides a better understanding of aspirin biodisposition in

TABLE 1 | Physiological changes in pregnancy that affect drug pharmacokinetics and pharmacodynamics.

System	Parameter	Change during pregnancy	References
Cardiovascular	Cardiac output	↑ up to 50%	[11]
	Stroke Volume	↑	[12]
	Heart Rate	↑ (by 10–20 bpm)	[13]
	Blood Pressure	↓ (slight, mid-pregnancy)	[13]
	Systemic Vascular Resistance	↓	[14]
Respiratory	Tidal Volume	↑ ~40%	[15]
	Total Lung Capacity	↓ ~4%	[16, 17]
Gastrointestinal	Hepatic Blood Flow	↑ up to 60%	[18]
	Motility	↓	[11]
	Gastric Emptying	↓	[18]
	Gastric pH	↓ (slightly more acidic)	[18]
Renal	Glomerular Filtration Rate (GFR)	↑ ~50%	[19]
	Renal Plasma Flow	↑ 60%–80%	[20]
Hematological	Plasma Volume	↑ ~45%	[11]
	Plasma Protein Binding	↓ (albumin, α 1-acid glycoprotein)	[21]
	Red Blood Cell Mass	↑ 20%–30%	[12]

pregnancy, facilitating improved exposure–response assessments for safety and efficacy. By predicting both maternal and fetal PK throughout gestation, the model can help healthcare providers tailor aspirin therapy to minimize risks and maximize benefits, thereby improving outcomes for both mother and fetus.

2 | Materials and Methods

2.1 | Institutional Review Board Approval

This study used computational modeling and did not involve human subjects and did not require approval from the Institutional Review Board.

2.2 | Software Platform

PBPK modeling and simulations for aspirin were implemented using Gastroplus (version 9.9, Simulation Plus Inc., Lancaster, CA, USA). PBPK simulations were developed using drug-dependent parameters (Absorption, Distribution, Metabolism, Excretion and physicochemical properties) (Figure 1A–D) from ADMET Predictor (version 11, Simulation Plus Inc., Lancaster, CA) and the literature. Population-dependent physiological parameters in models were obtained using the Population Estimates for Age-Related Physiology (Simulation Plus Inc., Lancaster, CA, USA).

2.3 | Model Development

The models were developed using Gastroplus and ADMET Predictor in the non-pregnant population, then validated (see

below; from the non-enteric-coated data) before being developed in the pregnant population.

Each model was initially developed using the pharmacokinetic, physiochemical, or biopharmaceutical parameters obtained from ADMET Predictor and the literature (Figure 1A). Pharmacokinetic profiles of aspirin in both non-pregnant and pregnant women were obtained from the literature [22, 23, 25], scanned with GetData Graph Digitizer version 2.26, and incorporated into the model.

Initially, single-batch simulations were conducted to compare observed and predicted values. When deviations between predicted and observed values occurred, the model underwent refinement through parameter optimization using the published data. Population simulations were initiated once discrepancies were minimized, with observed and predicted values aligning within 90% in single-batch simulations.

Population simulations involved 100 virtual subjects for each group: nonpregnant, first-trimester pregnant, second-trimester pregnant, and third-trimester pregnant. Subjects received 100 mg non-enteric aspirin in the pregnant population and 81 mg non-enteric coated aspirin in the nonpregnant population [26]. Based on these 100 subjects, the population-simulated mean values of pharmacokinetic parameters were derived.

2.4 | Aspirin and Salicylic Acid Input Parameters

The absorption, metabolism, distribution, elimination (ADME), and physicochemical properties of aspirin and salicylic acid were obtained from literature or generated within the ADMET

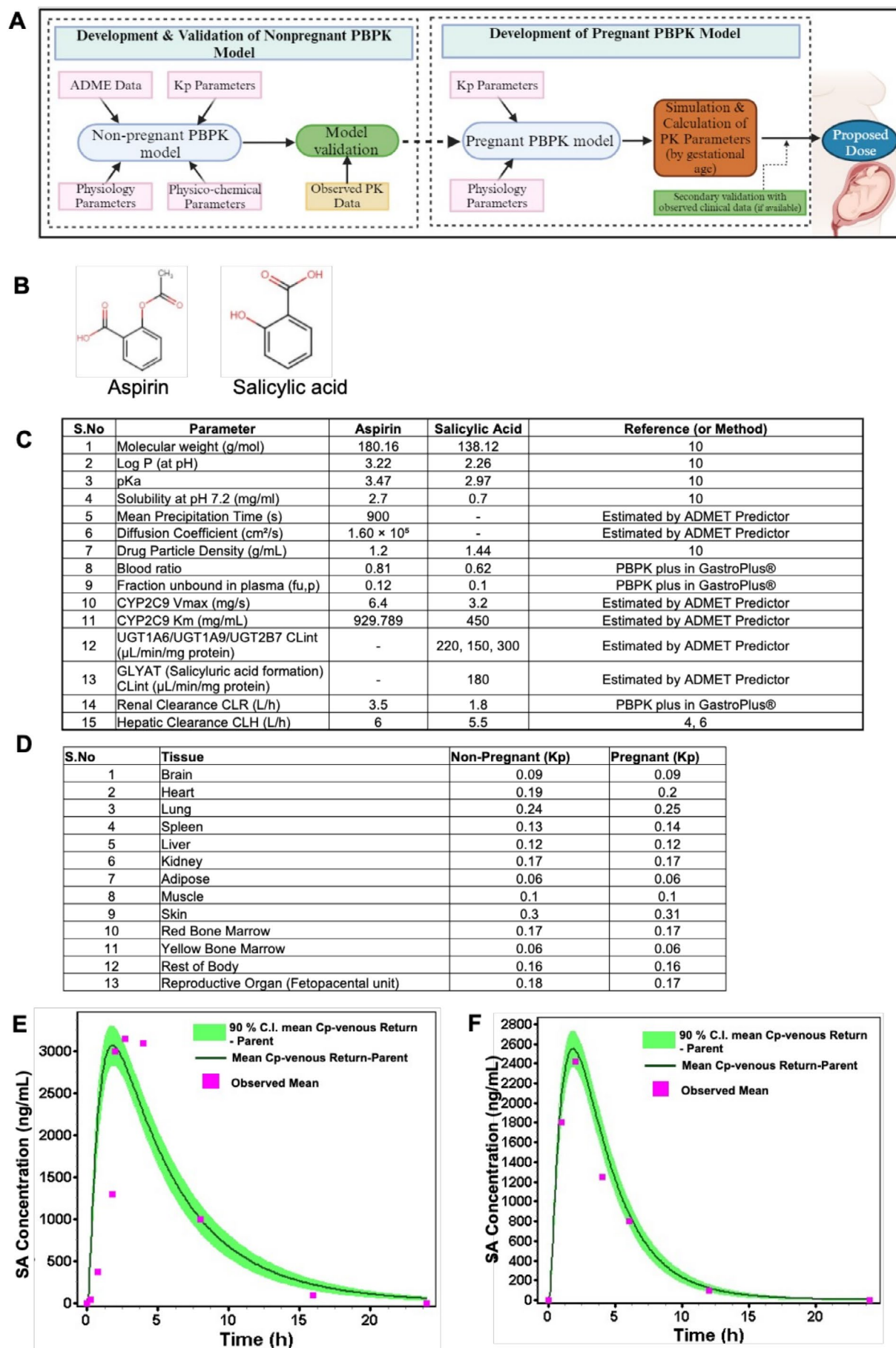


FIGURE 1 | Legend on next page.

FIGURE 1 | Development & Validation of a Physiologically Based Pharmacokinetic (PBPK) Model. (A) Workflow of developing a pregnancy PBPK in Gastroplus. (B) Chemical structure of aspirin and its active metabolite salicylic acid. (C) Physicochemical properties of aspirin and salicylic acid. (D) Tissue-to-Plasma Partition Coefficients (Kp) of Aspirin. Simulated and observed plasma concentration vs. time profiles in (E) non-pregnant individuals and (F) pregnant individuals, where simulated data are displayed as mean(solid black line) and 90% simulation interval (green, $N=100$), and observed data are means(pink squares, $N=20$, Patrick et al. and Shanmugan et al.).

TABLE 2 | Observed and predicted PK parameters after oral dosing of aspirin.

Pharmacokinetic parameter	Nonpregnant			Pregnant		
	Observed ^a [25]	Predicted	Fold error	Observed ^b [8]	Predicted	Fold error
C_{max} (mg/mL)	3150	3161.4	1.01	2417	2501.7	1.04
T_{max} (h)	2.68	1.936	0.72	2	1.8879	0.94
AUC_y (mg×h/mL)	2.07×10^4	2.01×10^4	0.97	Not listed	12,090	—
AUC_t (mg×h/mL)	2.06×10^4	1.615×10^4	0.78	12187.3	12,050	0.99
Clearance (L/h)	3.922	3.5	0.89	6	5.25	0.88

Note: Reported values in nonpregnant subjects receiving a single oral dose of 81 mg of non-enteric coated aspirin.
^aReported values in pregnant subjects receiving a single oral dose of 100 mg of non-enteric coated aspirin.
^bA fold error (predicted/observed) of less than 1.5 indicates that model was able to adequately predict the pharmacokinetics of aspirin in pregnancy.

Predictor in Gastroplus. The input parameters for aspirin are presented in Figure 1C. Tissue-to-plasma partition coefficients of aspirin used in both the non-pregnant and pregnant PBPK models are presented in Figure 1D.

The PBPK model focuses on salicylic acid, the major, pharmacologically active metabolite of aspirin, which is generated rapidly by hydrolysis and plasma esterases and is further metabolized by acyl-CoA-acyltransferase (ACNAT), uridine 5'-diphosphoglucuronosyltransferases (UGTs), and CYP450s (CYP2C9, CYP3A4, CYP2E1) [27, 28]. Studies performed in the non-pregnant and pregnant populations obtained renal clearance (CLR) of salicylic acid [9, 23, 25, 29]. Additional details about aspirin input parameters can be found in the supplemental section.

2.5 | Validation of Models in Non-Pregnant & Pregnant Subjects

Both the nonpregnant and pregnant models were validated independently using the population simulation within Gastroplus with 100 subjects. PBPK-based predicted plasma concentration-time profiles were compared with clinically observed PK profiles for validation. Models were determined valid if the observed plasma concentration-time data fell within the 90% simulation interval of the simulated concentration-time profiles and all other pharmacokinetic properties had a fold error of less than 1.5 when comparing simulated to observed parameters [30].

2.6 | Statistical Analysis

Data were analyzed using Prism 7 (GraphPad, La Jolla, CA, USA) software. The Shapiro–Wilk test was conducted to check for the data’s normality. Ordinary one-way analysis of variance followed by multiple comparison tests was used to

compare normally distributed data with at least three means. The Kruskal-Wallis test with Dunn’s multiple comparison tests was used for the data that were not normally distributed.

3 | Results

3.1 | Prediction of Salicylic Acid in Nonpregnant & Pregnant Subjects

After optimizing relevant parameters, the clinically observed plasma concentration-time profiles of salicylic acid in pregnant and nonpregnant people were within the 5th to 95th percentile of predicted plasma concentration-time profiles within Gastroplus (Figure 1E,F).

The fold errors of all PK parameters for nonpregnant and pregnant people were ≤ 1.5 (Table 2), indicating that the pregnancy PBPK model can adequately predict the PK and PD of salicylic acid in pregnant women.

3.2 | Effect of Pregnancy Gestation Period on the Pharmacokinetics of Aspirin

Figure 2 shows plasma concentration-time profiles of aspirin in nonpregnant and pregnant women across all trimesters of pregnancy. The predicted systemic exposures (AUC_{0-24h}) to SA were reduced throughout gestation (by approximately 20% at 10weeks gestation, 25% at 20weeks gestation, 29% at 30weeks gestation, and 30% at 40weeks gestation) (Table 3). Additionally, higher clearance (5.7L/h from 10weeks of gestation to 5.9L/h at 40weeks of gestation) was noted as gestation progressed (Table 3). The increased clearance observed during pregnancy may be due to enhanced renal elimination and upregulated hepatic metabolism.

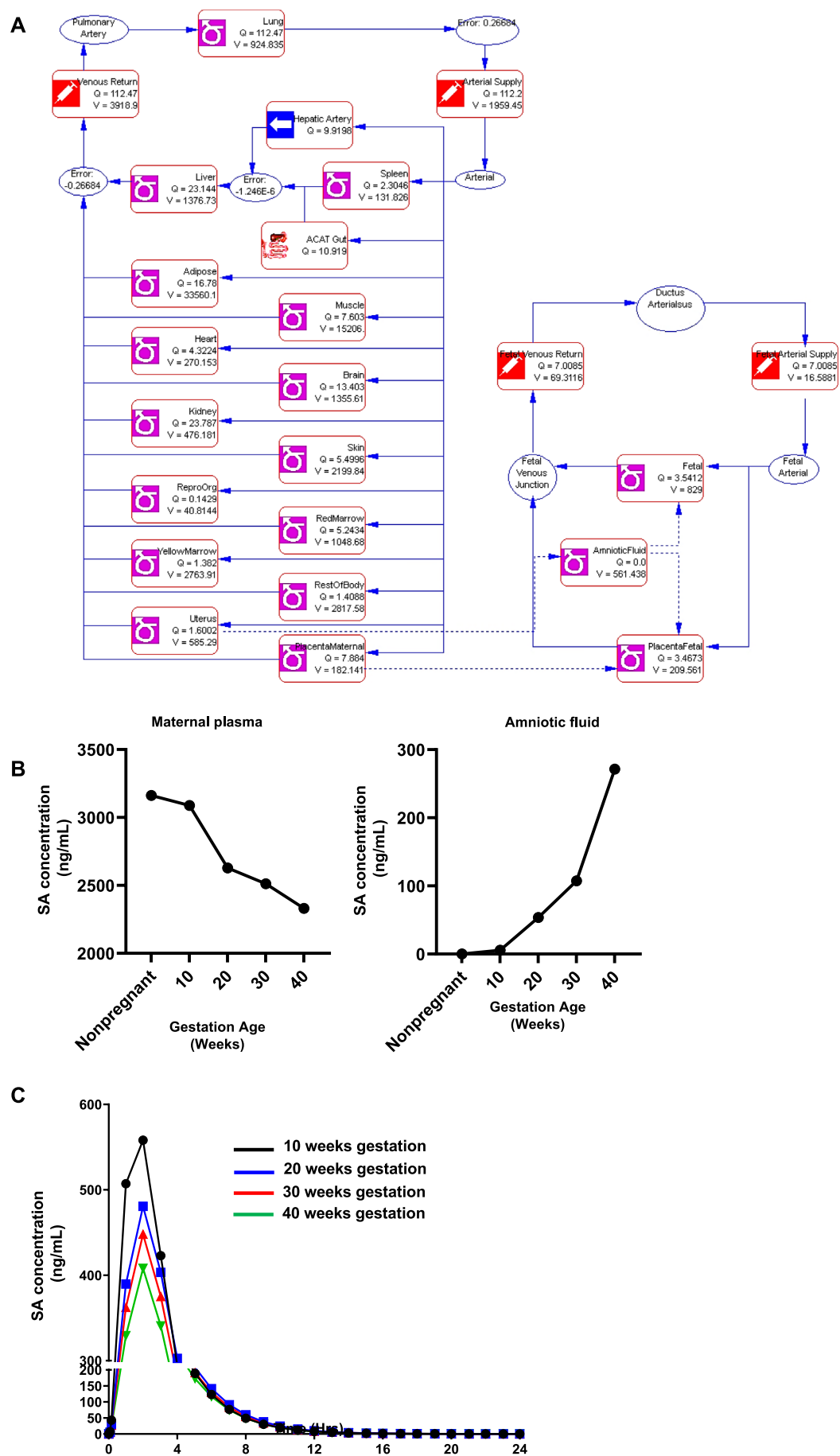


FIGURE 2 | Legend on next page.

FIGURE 2 | Simulated pregnant pharmacokinetics of aspirin. (A) Structure of PBPK model within Gastroplus (with associated volumes and flow rates). (B) Predicted pharmacokinetics of aspirin metabolite (SA) concentrations in maternal plasma, amniotic fluid, and fetal plasma. (C) Predicted concentration vs. time of aspirin metabolite (SA) in placental tissue.

TABLE 3 | Predicted maternal plasma PK parameters after oral dosing of aspirin across varying gestations of pregnancy.

Pharmacokinetic parameter	Gestational age of pregnancy			
	10 weeks	20 weeks	30 weeks	40 weeks
Clearance (L/h)	4.791	5.048	5.403	5.907
Vss (mg)/(ng/mL)	9.482	10.575	11.295	12.143
T _{1/2} (h)	1.44	1.452	1.449	1.424
C _{max} (mg/mL)	3088.7	2628.3	2512.4	2331.2
AUC _∞ (mg × h/mL)	1.295 × 10 ⁴	1.209 × 10 ⁴	1.155 × 10 ⁴	1.12 × 10 ⁴
AUC _t (mg × h/mL)	1.294 × 10 ⁴	1.207 × 10 ⁴	1.154 × 10 ⁴	1.118 × 10 ⁴
T _{max} (h)	1.57	1.8951	1.816	1.8792

TABLE 4 | Predicated fetal plasma PK parameters through gestation.

Pharmacokinetic parameter	Fetal plasma			
	10 weeks	20 weeks	30 weeks	40 weeks
C _{max} (mg/mL)	292.03	325.4	410.4	507.3
AUC _t (mg/mL × h)	1210.917	1488.95	1893.875	2300.4
T _{max} (h)	1.6	1.92	1.92	1.92
Fetal/maternal C _{max} ratio	0.1	0.12	0.16	0.22
Fetal/maternal AUC _t ratio	0.094	0.12	0.16	0.21

The change in systemic exposures to SA stratified by gestational age was minimal at 10% from 10 to 40 weeks' gestation. The change in clearance of aspirin stratified by gestational age was minimal at 3% from 10 to 40 weeks gestation (Table 3). The results suggest that the gestational changes in volume of distribution contribute to increased clearance, reduced maximum concentration (C_{max}), and reduced systemic exposures (AUC_{0-24h}) during pregnancy. The PBPK model predicted a reduction in SA exposure of ~20%–30% during pregnancy, corresponding to pregnant/nonpregnant AUC ratios of 0.72–0.79. These predictions align closely with published clinical data, which report observed ratios of approximately 0.7–0.8. This comparison demonstrates that the model successfully reproduced the exposure differences between pregnant and nonpregnant populations.

3.3 | Fetal Pharmacokinetics of Aspirin

There is very limited data on SA concentrations in the fetus after aspirin administration. The limited literature suggests that the concentration of aspirin in cord blood samples at delivery (aspirin was ingested by the mother before delivery when the cord was clamped) is time-dependent, varying from 66.8 mg/L at 12 min after delivery to 54.9 mg/L at 123 min after delivery in term infants after mothers received a one-gram intravenous bolus of aspirin in the first stage of labor [31]. The C_{max} ratio (fetal: maternal) ranged

from 0.02 to 0.99, indicating a high degree of participant variability. The pregnancy PBPK model predicted concentrations of C_{max} ratios within the literature-reported values.

As gestation progressed, both fetal-to-maternal plasma ratios and fetal plasma concentrations increased (Table 4). The predicted fetal plasma exposures and fetal to maternal plasma exposure ratios (AUC ratio: 0.094–0.21 and C_{max} ratio: 0.10–0.22) increased from 10 weeks' gestation to 40 weeks' gestation (Table 4). Not only did fetal plasma exposures increase throughout gestation, but so did amniotic fluid concentration, which sharply increased after 30 weeks' gestation.

In contrast, SA placental tissue concentrations significantly differed between 10 weeks' gestation and 40 weeks' gestation (Figure 2c). As gestation progressed, placental tissue concentrations decreased.

4 | Comment

4.1 | Principal Findings

Our study revealed that gestational age significantly affects the pharmacokinetics of aspirin in both maternal and fetal systems. Despite the absence of a dedicated pregnancy PBPK model for aspirin, our findings underscore the importance of

such models for predicting fetal exposure. Existing clinical studies on aspirin use during pregnancy exhibit considerable variability in dosage recommendations, and many obstetrical governing bodies offer conflicting guidelines on optimal aspirin dosages for pregnant women. This disparity highlights the urgent need for a pregnancy-specific PBPK model to predict fetal drug exposure and guide safer dosing protocols during pregnancy.

Gastroplus, a PBPK modeling and simulation tool, was utilized in this study to assess fetal and maternal PK of aspirin at different gestational ages of pregnancy. The pregnancy PBPK model created in Gastroplus accounted for anatomical and physiological changes during pregnancy. The models for pregnancy and nonpregnancy were validated using clinical data from the literature.

Within the pregnancy PBPK model, pharmacokinetic parameters indicate that aspirin exhibits approximately 14%, 14%, 15%, and 17% higher clearance and 20%, 25%, 29% and 30% lower exposures in pregnant women at 10, 20, 30, and 40 weeks' gestation of pregnancy, respectively compared to nonpregnant women. These findings indicate a potential requirement for higher aspirin dosages as pregnancy progresses.

The umbilical cord to maternal plasma ratios of aspirin reported in the literature vary from 0.02 to 0.9 in a single dose of intravenous aspirin [22, 23]. The PBPK model predicted that the AUC_{0-6h} fetal to maternal plasma ratio of SA was 0.09 at 10 weeks, 0.12 at 20 weeks, 0.16 at 30 weeks, and 0.21 at 40 weeks' gestation. The predicted C_{max} fetal to maternal ratio of SA was 0.10 at 10 weeks' gestation, 0.12 at 20 weeks' gestation, 0.16 at 30 weeks' gestation, and 0.22 at 40 weeks' gestation. The C_{max} ratios predicted in this model fell within the range reported in the literature. However, the ratios pertained to intravenously administered aspirin, whereas the PBPK model was developed based on oral administration. Since oral aspirin undergoes extensive first-pass metabolism with an overall bioavailability of only ~40%–50%, systemic exposures predicted by the oral model are expected to be lower than those observed with I.V. dosing, and this distinction should be considered when interpreting the results.

As the pregnancy progresses, the fetal AUC, C_{max} , and fetal-to-maternal ratios increase along with the fetal plasma concentration. However, the maternal plasma concentration decreases with progressing gestation. The physiological changes occurring during pregnancy could partly explain the lower maternal plasma concentration and higher fetal plasma concentrations (Figure 2). However, it is worth noting that as gestation progresses, there is a decrease in both the uterine and placental concentration of salicylic acid (Figure 2).

The PBPK model predicted appreciable concentrations of aspirin in amniotic fluid, with a significant increase in the concentration of aspirin in amniotic fluid from 30 to 40 weeks' gestation. There are no clinical studies published to date observing aspirin concentration in amniotic fluid at different gestational ages. There have been reports on the concentration of aspirin in amniotic fluid at 32 weeks' gestation related to the maternal dosage. In this study, as maternal plasma

concentration increased, SA concentrations within the amniotic fluid proportionately increased [23]. The model predictions are consistent with data reported that showed prolonged exposure to aspirin leading to increased fetal plasma and amniotic fluid concentrations.

The predicted SA concentrations in uterine tissue are approximately 2–3 times higher compared to fetal plasma (data not shown). Although the predicted fetal plasma exposures are approximately 90% higher at 40 weeks' gestation compared to 10 weeks' gestation of pregnancy, there was an appreciable difference in the uterine tissue exposure to aspirin observed throughout pregnancy. The increasing fetal exposure may be of concern in neonates who need invasive procedures completed shortly after delivery [32]. Additional clinical studies are necessary to confirm the predicted concentrations of SA in the placental tissues.

Many clinical studies have been conducted on aspirin use and the prevention of hypertensive disorders in pregnancy. Despite these numerous studies, very few studies have been completed looking at PK fetal exposure, and fetal to maternal concentrations [7–9]. As physicians are debating the best dosage of aspirin for the reduction of maternal hypertensive disorders of pregnancy, this PBPK model may provide additional guidance.

Therefore, the impact of predicted concentrations of aspirin during pregnancy and expected inhibition of thromboxane and prostaglandin production was reviewed. Currently there have been no single dose-finding studies in pregnancy conducted pertaining to the necessary dosage to inhibit thromboxane and prostaglandin production across various gestational ages. A study conducted by Caron et al. examined the use of 81 mg of aspirin and platelet function, and found that 81 mg was not enough to facilitate platelet function changes but did not propose a dosage at which it would affect platelet function [24].

Studies outside of pregnancy have suggested dosages of 81–325 mg are needed to reduce cardiovascular disease [25]. A study conducted by Patrick et al. examined both immediate and extended-release formulations of aspirin related to thromboxane inhibition. Both formulations produced dose-dependent inhibition. However, the dose required to maintain 50% of maximum serum thromboxane B2 with extended-release aspirin was 50 mg versus 30 mg for instant-release (non-enteric-coated) [25]. A study conducted by Sibai et al. utilized varying dosages of aspirin from 20 mg to 80 mg in women at term (37 weeks) to evaluate the levels of serum thromboxane inhibition. It was suggested that both 60 and 80 mg of aspirin provided thromboxane inhibition [26].

As the PK of SA changes during pregnancy, this PBPK model predicted a 20%–30% reduction in SA exposure in pregnant women compared to nonpregnant women, corresponding to predicted pregnant/nonpregnant AUC ratios of 0.72–0.79. These predictions closely align with observed clinical ratios of ~0.7–0.8 reported in the literature, indicating that the model successfully reproduced the clinically observed exposure differences and may be useful for individualizing aspirin dosages throughout pregnancy once a range of therapeutic concentrations has been established.

4.2 | Results in Context of What Is Known

Studies pertaining to the PK of aspirin in pregnancy are limited. We were able to compare our simulated PBPK results to published studies. Within our PBPK model, all PK parameters were within 1.5-fold errors, demonstrating an acceptable PBPK model for aspirin [30]. Within our model, we were able to adjust maternal body mass index (BMI), gestational age, maternal race, fetal sex, and other variables associated with alterations in PK.

4.3 | Clinical Implications

The results from our study may have important implications for clinical practice. Within this simulation model, examining the PK of aspirin across gestation, we identified reduced maternal plasma concentration as gestation progressed. Previous research has demonstrated the clinical implications between plasma exposure and physiologic response [7–9]. Although more work is needed to determine optimal salicylic acid exposures for safety and efficacy, this PBPK model further supports the current debate that optimal aspirin dosage may vary throughout pregnancy.

4.4 | Research Implications

The results from our study have important implications for research. This study highlights the use of technology to study medication PK within a therapeutically orphaned population. Currently, pregnant women have often been described as ‘therapeutic orphans’; however, there is now a worldwide effort by regulatory agencies, healthcare professionals, and other stakeholders to address these knowledge gaps and enhance the inclusion of pregnant women in clinical trials where feasible. This approach also complies with the FDA Modernization Act 2.0, which encourages the use of non-animal alternative methods, including PBPK modeling, to evaluate drug safety and efficacy in vulnerable populations. By providing a rigorous, mechanistic framework to predict maternal and fetal drug disposition, our model supports this initiative while minimizing the need to expose pregnant women to unnecessary clinical risk.

Additionally, this study addressed the need for the development of model simulations to inform aspirin dosage in pregnancy. Physiological changes throughout pregnancy and individual variability illustrate one of the main reasons for developing a PBPK model to identify optimal dosages for preeclampsia risk reduction.

4.5 | Strengths & Limitations

4.5.1 | Strengths

This study represents the first attempt to develop a physiologically based pharmacokinetic (PBPK) model for aspirin (ASA) and salicylic acid (SA) during pregnancy. We utilized existing pharmacokinetic (PK) data from both pregnant and non-pregnant individuals, allowing us to validate the model against observed data for both enteric-coated (EC) and non-EC formulations. Additionally, the model successfully predicts theoretical

concentration-time profiles for ASA/SA in both maternal and fetal tissues, paving the way for future experimental validation. Another significant strength is that the model enables the exploration of how demographic and physiological variations influence the pharmacokinetics of ASA/SA, providing valuable insights for personalized drug use during pregnancy.

4.5.2 | Limitations

Despite these strengths, the model has several limitations. Data availability for fetal ASA/SA concentrations is limited, which hinders thorough validation of fetal predictions. Moreover, the process of parameter fitting and optimization within the PBPK model carries the risk of overfitting, where seemingly good model performance may still incorporate inappropriate parameters or miss key contributing factors, such as changes in metabolic enzymes or transporters. Additionally, there remains a lack of comprehensive understanding of the exposure-response (E-R) relationship for salicylic acid in pregnancy, particularly concerning its safety and efficacy, underscoring the need for further data collection and model refinement.

5 | Conclusion

A PBPK model was successfully developed to predict pregnancy-related PK changes in pregnancy seen with oral non-enteric coated aspirin that agreed with clinical data. The simulated PK in this model is like observations in non-pregnant and pregnant individuals and demonstrates that systemic salicylic acid exposures are reduced by approximately 20%–30% in pregnant women compared to nonpregnant women.

The predicted fetal plasma exposures to aspirin increased by approximately 90% and the fetal to maternal plasma exposure ratio increased from 0.1 to 0.22 from 10 weeks’ gestation to 40 weeks’ gestation of pregnancy. The higher fetal exposures to aspirin during the final weeks of pregnancy should be verified in future, prospective studies.

Additional clinical studies are warranted to adequately evaluate the therapeutic effect vs. exposure vs. dose relationship of aspirin in pregnant women to determine if dosage changes by gestational age are warranted for clinical efficacy in reducing the incidence of hypertensive disorders of pregnancy.

Author Contributions

A.C.-S., A.K.K., M.P., M.M.C., and R.M. wrote the manuscript; A.K.K. and X.M.W. designed the research; A.C.-S. and A.K.K. performed the research; and A.C.-S. analyzed the data.

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Conflicts of Interest

The authors declare no conflicts of interest.

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